Large-Model Prior and Causal Consistency for Unseen-Condition Multimodal Defect Recognition in SLM

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***Abstract*—Selective Laser Melting (SLM) monitoring provides multi-view RGB and infrared observations, but defect recognition becomes unreliable under unseen conditions. This paper studies a setting where `data1 official_cv` holds out complete `condition_uids` and `data2 transfer_test` serves as a harder external transfer benchmark. The problem is how to use a large-model prior for small-sample multimodal SLM recognition without overfitting to condition-specific shortcuts. Direct prompt-based adaptation of Qwen2-VL reaches only 0.3914 Macro-F1 on the 3-class `data1 official_cv` task. We therefore reuse the Qwen2-VL vision tower as a pretrained visual prior, fuse multimodal views with an explicit MLP head, and impose a cross-condition causal consistency loss on the fused representation. The main regime keeps the visual prior frozen and trains only the lightweight head. On `data1 official_cv`, the proposed method reaches 0.8547 Macro-F1 and 0.8556 balanced accuracy, outperforming prompt-Qwen and the strongest classical baseline. On `data2 transfer_test`, it reaches 0.6067 Macro-F1. Input ablations show that RGB view 1, RGB view 2, and infrared observations all contribute useful information, although the current Qwen branch is best instantiated with dual RGB. Additional experiments show that true fine-tuning of the last two visual blocks is harmful, indicating that preserving the large-model prior and enforcing causal consistency is more effective than aggressive end-to-end adaptation.**

***Index Terms*—Selective laser melting, multimodal defect recognition, unseen-condition generalization, external transfer, large-model prior, causal consistency, Qwen2-VL.**

I. Introduction

SELECTIVE Laser Melting (SLM) is a representative metal additive manufacturing process whose final part quality is strongly affected by melt-pool stability, thermal diffusion, and layer-wise geometric evolution. Practical monitoring systems therefore collect multiple sensing views, especially optical images and infrared observations, in order to infer defect states during printing [1]–[3]. For industrial deployment, however, the central challenge is not merely whether a classifier works under a random split. The more relevant question is whether multimodal defect recognition remains reliable under unseen process conditions.

The repository studied in this paper directly targets that question. Its main benchmark, `data1 official_cv`, splits samples by `condition_uid` so that train and test sets do not share the same process condition. A second benchmark, `data2 transfer_test`, is reserved as a harder external transfer stress test involving unseen models, unseen process settings, and broader distribution shift. This dual evaluation protocol makes the project suitable for studying unseen-condition multimodal recognition rather than ordinary in-domain classification.

Large vision-language models provide a natural starting point because they encode strong pretrained visual priors. Qwen2-VL is particularly attractive due to its high-capacity visual encoder and flexible multimodal interface [6]. Yet the repository history also reveals a negative result: directly adapting Qwen2-VL through prompt-based closed-set scoring performs poorly on the current industrial task. This suggests that the limitation is not the absence of a large-model prior itself, but the way that prior is transferred to a small-sample and condition-shifted recognition problem.

This paper therefore reframes the problem as discriminative adaptation of a large-model prior. The proposed method keeps Qwen2-VL as the visual foundation, explicitly fuses multimodal observations, and imposes a causal consistency regularizer on same-label samples drawn from different conditions. The main hypothesis is that defect-relevant evidence should remain stable across conditions, while textures, illumination patterns, and thermal backgrounds may vary. Under this view, causal consistency is not used to claim a full causal graph of SLM, but to bias the learned representation toward condition-stable defect cues. The design also keeps multimodal evidence central throughout the paper: the best Qwen configuration currently uses dual RGB input, yet infrared remains meaningful because IR-only recognition is non-trivial and triple-view transfer is strong in the classical baseline family.

The contributions are summarized as follows.

1) An unseen-condition evaluation narrative is formalized for multimodal SLM defect recognition, where `data1 official_cv` is the source-side condition-heldout benchmark and `data2 transfer_test` is the external transfer benchmark.
2) A four-stage framework is developed around a Qwen2-VL large-model prior: multimodal feature extraction, causal consistency representation regularization, lightweight large-model adaptation, and closed-set prediction output.
3) Experiments show that causal consistency improves small-sample adaptation of the large-model prior, while direct prompt-based adaptation and aggressive true fine-tuning both underperform under unseen-condition and transfer evaluation.

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II. Related Work

A. Multimodal Monitoring and Defect Recognition in SLM

SLM and powder-bed-fusion monitoring have been extensively studied through optical images, infrared thermography, acoustic signals, and multi-sensor combinations [1]–[5]. Optical monitoring is effective for observing surface irregularities and recoating artifacts, while infrared imaging captures thermal signatures associated with subsurface instability and lack-of-fusion behavior. Recent multisensor studies further show that complementary modalities can improve process understanding when the fusion mechanism is aligned with the sensing physics. These findings motivate this paper to treat RGB view 1, RGB view 2, and IR as meaningful defect sources rather than interchangeable inputs.

B. Large-Model Adaptation Under Distribution Shift

Large pretrained vision-language models provide transferable visual priors, but strong pretraining does not guarantee robustness under industrial condition shift. Qwen2-VL is a recent strong example of high-capacity visual-language pretraining [6]. Meanwhile, invariant learning and robust optimization literature emphasizes that models should be guided away from environment-specific shortcuts when test environments differ from training ones [7], [8]. This perspective is directly relevant here: small-sample industrial adaptation can easily overfit to condition-dependent textures or thermal backgrounds, so the key problem is how to preserve the large-model prior while encouraging condition-stable representations.

III. Unseen-Condition Task Definition and Evaluation Protocols

A. Source-Side Condition-Heldout Benchmark

The main benchmark is the 3-fold `data1 official_cv` protocol. Instead of randomly splitting images, the repository splits by `condition_uid`, ensuring that all samples from the same process condition remain in the same fold. This design prevents leakage of highly similar background patterns, layer textures, and thermal statistics across train and test sets. The primary task is 3-class defect recognition with labels `normal`, `HEW`, and `LEL`; `HEV` is merged into `HEW` in the released manifests.

B. External Transfer Benchmark

The `data2 transfer_test` set is not used for model selection. Instead, it serves as an external transfer test with larger distribution shift involving unseen models, unseen process settings, and broader role variation. This distinction is important for interpretation: absolute scores are expected to be lower than on `data1`, but relative performance under `data2` more directly reflects practical robustness.

C. Multimodal Inputs and Evaluation Targets

The repository supports four input modes: `rgb_view1`, `rgb_dual`, `ir_only`, and `triple_view`. Throughout the paper, these are viewed as different instantiations of the modality set $\mathcal{M} \subseteq \{\mathrm{rgb1}, \mathrm{rgb2}, \mathrm{ir}\}$. The main paper target is 3-class recognition because the Qwen branch is substantially more stable on 3-class learning than on direct 2-class training.

TABLE I
Statistics of the Source and Transfer Protocols

Domain	Samples	Conditions	Normal	HEW	LEL
`data1`	289	22	104	86	99
`data2`	404	22	167	128	109

Fig. 1. Protocol overview of `data1 official_cv` and `data2 transfer_test`. Placeholder for the final figure.

Table I summarizes the two protocols. Fig. 1 is reserved for the final visualization of the condition-heldout source split and the harder transfer benchmark.

IV. Method

A. Multimodal Data Feature Extraction

For sample i, let $\mathcal{M}_i = \{m_1, \ldots, m_{K_i}\}$ denote the available modalities, where each m corresponds to one of $\{$`rgb_view1`, `rgb_view2`, `ir`$\}$. The pretrained Qwen2-VL visual encoder f_{θ_v} is used as the large-model prior. For modality $m \in \mathcal{M}_i$, the encoder outputs a sequence of visual tokens:

$$\mathbf{T}_i^{(m)} = f_{\theta_v}\left(\mathbf{x}_i^{(m)}\right) \in \mathbb{R}^{P_i^{(m)} \times d}, \tag{1}$$

where $P_i^{(m)}$ is the number of visual tokens after Qwen2-VL spatial merging and d is the hidden size. Following the repository implementation, the token sequence is pooled by simple averaging:

$$\mathbf{z}_i^{(m)} = \frac{1}{P_i^{(m)}} \sum_{p=1}^{P_i^{(m)}} \mathbf{T}_{i,p}^{(m)} \in \mathbb{R}^d. \tag{2}$$

The modality vectors are concatenated and passed through a lightweight fusion head:

$$\mathbf{g}_i = \mathrm{Concat}\left(\mathbf{z}_i^{(m_1)}, \ldots, \mathbf{z}_i^{(m_{K_i})}\right), \tag{3}$$

$$\tilde{\mathbf{h}}_i = \phi(\mathrm{LN}(\mathbf{g}_i)), \quad \mathbf{r}_i = \frac{\tilde{\mathbf{h}}_i}{\|\tilde{\mathbf{h}}_i\|_2}, \tag{4}$$

where $\phi(\cdot)$ denotes the two-layer MLP fusion head with GELU and dropout. The main official run uses $K_i = 2$ with dual RGB views. This should not be interpreted as evidence that IR is unhelpful; rather, it shows that the current Qwen fusion head extracts the most stable gain from the dual RGB path, while IR still carries complementary thermal information.

B. Causal Consistency Representation Constraint

Let y_i denote the defect label and c_i the process condition. For each training sample i, the dataset constructs a positive pool

$$\Omega_i = \{j \mid j \neq i,\ y_j = y_i,\ c_j \neq c_i\}, \tag{5}$$

which contains same-label samples from different conditions. During training, one positive index $\pi(i)$ is sampled from Ω_i when the pool is non-empty. For a mini-batch $\mathcal{B}$, define

$$\mathcal{B}^+ = \{i \in \mathcal{B} \mid \Omega_i \neq \varnothing\}. \tag{6}$$

The causal consistency loss is then written as

$$\mathcal{L}_{\mathrm{cc}} = \frac{1}{|\mathcal{B}^+|} \sum_{i \in \mathcal{B}^+} \left(1 - \mathbf{r}_i^\top \,\mathrm{sg}\big(\mathbf{r}_{\pi(i)}\big)\right), \tag{7}$$

where $\mathrm{sg}(\cdot)$ denotes stop-gradient on the positive branch, matching the repository implementation in which the positive representation is detached for stability and memory efficiency. Because both vectors are ℓ_2-normalized, (7) is equivalent to minimizing cosine distance between same-label representations across different conditions.

The term "causal consistency" is used in an operational sense. The goal is not to recover a full causal graph of the SLM process, but to encode the assumption that the defect identity is a more stable explanatory factor than condition-specific visual shortcuts. Aligning same-label samples across different conditions therefore encourages the model to retain defect-relevant evidence while suppressing unstable condition cues.

C. Large-Model Adaptation Strategy

The trainable parameter set depends on the adaptation regime:

$$\Theta_{\mathrm{train}} = \begin{cases} \{\theta_f, \theta_c\}, & \text{lightweight adaptation,} \\ \{\theta_v^{\mathrm{last2}}, \theta_m, \theta_f, \theta_c\}, & \text{true fine-tuning,} \end{cases} \tag{8}$$

where θ_f denotes the MLP fusion head, θ_c the classifier, $\theta_v^{\mathrm{last2}}$ the last two visual transformer blocks of Qwen2-VL, and θ_m the Qwen visual merger. The lightweight regime is the proposed main path because it preserves the large-model prior and reduces overfitting risk. The true fine-tuning regime is a diagnostic comparison that tests whether further visual-side adaptation is beneficial in the current small-sample setting.

D. Output Layer, Objective, and Pseudo-Code

Closed-set prediction is produced from the unnormalized fused representation:

$$\boldsymbol{\ell}_i = \mathbf{W}_c \tilde{\mathbf{h}}_i + \mathbf{b}_c, \qquad \hat{\mathbf{y}}_i = \mathrm{softmax}(\boldsymbol{\ell}_i). \tag{9}$$

The classification loss is

$$\mathcal{L}_{\mathrm{cls}} = -\frac{1}{|\mathcal{B}|} \sum_{i \in \mathcal{B}} \log \hat{y}_{i,y_i}, \tag{10}$$

and the overall objective becomes

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \lambda \mathcal{L}_{\mathrm{cc}}, \tag{11}$$

where $\lambda = 0.1$ in the official causal configuration. The full training logic is summarized in Table II.

TABLE II
PSEUDO-CODE OF THE PROPOSED TRAINING PROCEDURE

Require: Training set $\mathcal{D}$, pretrained Qwen visual encoder f_{θ_v}, adaptation regime, λ

Initialize fusion head θ_f and classifier θ_c
Freeze or unfreeze parameters according to (8)
for each epoch **do**
  for each mini-batch $\mathcal{B}$ **do**
    Encode each modality image with f_{θ_v} and mean-pool visual tokens
    Concatenate modality vectors and compute $\tilde{\mathbf{h}}_i$, $\mathbf{r}_i$, and $\hat{\mathbf{y}}_i$
    Sample $\pi(i)$ from Ω_i for each anchor with an available cross-condition positive
    Compute $\mathcal{L}_{\mathrm{cls}}$ and $\mathcal{L}_{\mathrm{cc}}$ using (7)
    Update the trainable parameters by minimizing (11)
  end for
  Select the best checkpoint by validation Macro-F1
end for
Evaluate the selected checkpoint on `data1 official_cv` and `data2 transfer_test`

V. Experiments and Analysis

A. Experimental Settings

The main Qwen configuration uses the Qwen2-VL-2B-Instruct vision tower, `rgb_dual` input, hidden dimension 512, dropout 0.1, batch size 4, AdamW optimization with learning rate 3×10^{-4} and weight decay 10^{-4}, cosine scheduling, mixed precision, and early stopping patience 2. The causal consistency weight is 0.1. The diagnostic true fine-tuning regime uses batch size 1, 3 epochs, early stopping patience 1, visual learning rate 2×10^{-6}, fusion-head learning rate 5×10^{-5}, and gradient clipping 0.5. The strongest classical baseline is ResNet18 with `rgb_dual` input and MLP-concat fusion, trained for 20 epochs. All source-side results are aggregated over three folds.

B. Main Results on the Source Benchmark

Table III reports the 3-class results on `data1 official_cv`. Prompt-Qwen reaches only 0.3914 Macro-F1, indicating that direct prompt-style closed-set adaptation fails to turn Qwen2-VL into a reliable unseen-condition recognizer. In contrast, once the same pretrained model is repurposed as a large-model visual prior inside an explicit fusion classifier, performance rises to 0.8402 Macro-F1. Adding causal consistency further improves the score to 0.8547, exceeding the strongest classical baseline at 0.8323.

The auxiliary 2-class evidence supports the same interpretation. Prompt-Qwen reaches only 0.3903 Macro-F1, while the causal Qwen fusion branch reaches 0.4707. Both are much weaker than the best classical 2-class baseline, which is why this paper treats 3-class recognition as the main task and uses direct 2-class learning only as a negative side observation.

C. Multimodal Effectiveness Under Unseen Conditions

Table IV studies modality combinations within the Qwen fusion family. Three findings are important. First, IR-only input still reaches 0.6270 Macro-F1, so thermal information is useful rather than redundant. Second, the dual RGB configuration is the strongest Qwen instantiation in the current implementation, indicating that complementary appearance

Fig. 2. Overall framework of the proposed method. The final figure should visualize four stages: multimodal feature extraction from RGB and IR observations, causal consistency regularization across unseen conditions, lightweight versus aggressive large-model adaptation, and closed-set prediction output.

TABLE III
MAIN 3-CLASS RESULTS ON `data1 official_cv`

Family	Setting	Accuracy	Balanced Accuracy	Macro-F1
Prompt-Qwen	3-class prompt baseline	0.4809 ± 0.0602	0.5021 ± 0.0460	0.3914 ± 0.0587
Classic	ResNet18 + `rgb_dual` + MLP-concat	0.8349 ± 0.0858	0.8412 ± 0.0846	0.8323 ± 0.0892
QwenVisFusion	Frozen-prior baseline	0.8383 ± 0.1009	0.8422 ± 0.1061	0.8402 ± 0.1048
QwenVisFusion	Frozen prior + causal consistency	**0.8521 ± 0.0929**	**0.8556 ± 0.0973**	**0.8547 ± 0.0955**
QwenVisFusion Last2 FT	Unfreeze last two visual blocks	0.7850 ± 0.0378	0.7979 ± 0.0332	0.7781 ± 0.0411

Fig. 3. Main 3-class comparison on `data1 official_cv`. Placeholder for the final figure.

TABLE IV
INPUT ABLATION OF QWENVISFUSION ON THE 3-CLASS SOURCE BENCHMARK

Input Mode	Balanced Accuracy	Macro-F1
`rgb_view1` baseline	0.7714 ± 0.0771	0.7419 ± 0.1203
`ir_only` baseline	0.6508 ± 0.0708	0.6270 ± 0.0878
`rgb_dual` baseline	0.8422 ± 0.1061	0.8402 ± 0.1048
`triple_view` baseline	0.8169 ± 0.0659	0.8121 ± 0.0615
`rgb_dual` + causal consistency	**0.8556 ± 0.0973**	**0.8547 ± 0.0955**

cues from two optical views are particularly stable under the present training regime. Third, directly adding IR to the dual RGB branch does not surpass dual RGB, which suggests that multimodal effectiveness depends on the fusion structure rather than on simply stacking more views.

This result should not be read as "IR is ineffective." Instead, it shows that all modalities are informative, but the current Qwen branch extracts the cleanest unseen-condition signal from the dual RGB pathway. This interpretation is reinforced by the classical transfer results, where the strongest external-transfer baseline uses triple-view late fusion rather than dual RGB.

D. Why Causal Consistency Helps and True Fine-Tuning Hurts

The gain from causal consistency is modest compared with the large jump from prompt-Qwen to explicit fusion, but it is the core methodological contribution of the paper. On `data1 official_cv`, causal consistency improves Macro-F1 from 0.8402 to 0.8547 and balanced accuracy from 0.8422 to 0.8556. This is precisely the evidence needed for the main claim: once the large-model prior has been converted into a discriminative multimodal recognizer, representation-level causal consistency provides additional robustness against unseen conditions.

The last-two-block experiment clarifies the boundary of that claim. If the final visual blocks are unfrozen, Macro-F1 drops from 0.8547 to 0.7781 on `data1 official_cv`; on `data2 transfer_test`, it drops from 0.6067 to 0.5808. The degradation indicates representation drift under limited industrial supervision. In other words, the challenge is not that the large-model prior is insufficiently adapted, but that aggressive true fine-tuning overfits condition-specific cues and weakens transfer.

E. External Transfer Results

Table V reports the best 3-class transfer results on `data2 transfer_test`. Prompt-Qwen reaches 0.4073 Macro-F1, the strongest classical transfer baseline reaches 0.4236, and the proposed causal QwenVisFusion reaches 0.6067. The ranking is important because `data2` combines multiple sources of unseen shift. Even though absolute scores are lower than on `data1`, the proposed method preserves the clearest robustness margin under the harder benchmark.

TABLE V
BEST 3-CLASS TRANSFER RESULTS ON `DATA2 TRANSFER_TEST`

Family / Best Model	Balanced Accuracy	Macro-F1
Prompt-Qwen / `data1_official_cv_3class_baseline`	0.4586 ± 0.0286	0.4073 ± 0.0511
Classic / triple-view late fusion	0.4973 ± 0.0954	0.4236 ± 0.1110
QwenVisFusion / `rgb_dual` + causal consistency	$\mathbf{0.6359 \pm 0.0158}$	$\mathbf{0.6067 \pm 0.0351}$

Fig. 4. Best-model comparison on `data2 transfer_test`. Placeholder for the final figure.

The transfer evidence also helps complete the multimodal story. Infrared remains meaningful in additive manufacturing monitoring and is still useful in the classical transfer family [4], [5]. The reason the best Qwen branch currently uses dual RGB is therefore methodological, not physical: the present fusion head has not yet extracted the full benefit of thermal information under limited industrial data.

VI. Conclusion

This paper studied unseen-condition multimodal defect recognition in SLM through the lens of large-model adaptation. The repository evidence shows that direct prompt-based Qwen2-VL adaptation is inadequate for the current small-sample industrial task. A stronger route is to keep Qwen2-VL as a large-model visual prior, perform explicit multimodal fusion, and impose causal consistency across same-label samples from different conditions. The resulting framework achieves the best source-side and external-transfer results in the repository. Additional experiments further show that aggressive unfreezing of the final visual blocks is harmful, which sharpens the main conclusion: in the present regime, transferability is improved more by preserving the pretrained prior and regularizing representations than by deeper end-to-end fine-tuning. Future work should focus on stronger infrared-aware fusion and more conservative visual-side adaptation mechanisms, such as adapters or LoRA-style modules, instead of direct block unfreezing.

References

[1] C. Chua, Y. Liu, R. J. Williams, C. K. Chua, and S. L. Sing, "In-process and post-process strategies for part quality assessment in metal powder bed fusion: A review," *Journal of Manufacturing Systems*, vol. 73, pp. 75–105, Apr. 2024, doi: 10.1016/j.jmsy.2024.01.004.

[2] M. Abdelrahman, E. W. Reutzel, A. R. Nassar, and T. L. Starr, "Flaw detection in powder bed fusion using optical imaging," *Additive Manufacturing*, vol. 15, pp. 1–11, May 2017, doi: 10.1016/j.addma.2017.02.001.

[3] J. L. Bartlett, F. M. Heim, Y. V. Murty, and X. Li, "In situ defect detection in selective laser melting via full-field infrared thermography," *Additive Manufacturing*, vol. 24, pp. 595–605, Dec. 2018, doi: 10.1016/j.addma.2018.10.045.

[4] J. Petrich, Z. Snow, D. Corbin, and E. W. Reutzel, "Multi-modal sensor fusion with machine learning for data-driven process monitoring for additive manufacturing," *Additive Manufacturing*, vol. 48, part B, Art. no. 102364, Dec. 2021, doi: 10.1016/j.addma.2021.102364.

[5] C. Maucher, K. T. Werkle, and H.-C. Möhring, "In-Situ defect detection and monitoring for laser powder bed fusion using a multi-sensor build platform," *Procedia CIRP*, vol. 104, pp. 146–151, 2021, doi: 10.1016/j.procir.2021.11.025.

[6] P. Wang *et al.*, "Qwen2-VL: Enhancing vision-language model's perception of the world at any resolution," arXiv:2409.12191, 2024, doi: 10.48550/arXiv.2409.12191.

[7] M. Arjovsky, L. Bottou, I. Gulrajani, and D. Lopez-Paz, "Invariant risk minimization," arXiv:1907.02893, 2019, doi: 10.48550/arXiv.1907.02893.

[8] S. Sagawa, P. W. Koh, T. B. Hashimoto, and P. Liang, "Distributionally robust neural networks for group shifts: On the importance of regularization for worst-case generalization," arXiv:1911.08731, 2019, doi: 10.48550/arXiv.1911.08731.